SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I L.Sathyam Sharma (192526O282) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L.Sathyam

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

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CLOUD COMPUTING & DATA ANALYSIS
CSA1519
CO2 - AT-3

1. Cloud - Based Big data Architecture for Retail :-

A retail company can use cloud services to collect and analyse customers transactions from online and offline stores

Architecture :- Data is collected using Apache kafka and Stored in large data lake. Apache spark spark processes large datasets and dashboards displays customers Purchasing patterns

Batch and steam processing :-
Batch Processing analyzes historical sales, while stream processing analyze live transaction for instant

Data lake and warehouse :-
A data lake stores raw data while a data warehouse stores cleaned and structural data for reporting.

scalability and cost :-
Auto - scaling data compression, Partitioning and pay as you pay - use Pricing reduce costs.

3

2. Secure cloud Architecture for Healthcare :-
A healthcare organization can store Patient records and medical images securely using a private cloud network and encrypted cloud storage

• Encryption :-
Encrypt data both at rest and during transmission. use a key management service to securely manage Encryption keys

identity and access Management (IAM)
use role based access control and multi - factor authentication (MFA) to ensure only authorized doctors access Patients records

Auditing and monitoring:
maintain logs of data access and changes use security monitoring tools to detect suspicious activities and support compliance

Preventing data breaches:-
Apply least - privilege access regular, security updates, backups, and network Firewalls

3 serverless cloud Architecture for Ticket Booking..

serverless computing allows applications to run without directly managing servers. cloud providers automatically manage the infrastructure

Architecture:-
API Orchestrator receives requests, serverless Functions process bookings and managed data bases store tickets details

Event-driven computing:-
Function automatically scale when booking requets increases

- Managed services :- Message queues manage requests during heavy traffic.

- Advantages & Limitages :- Serverless reduces server management and costs for variable workloads but could starts and Execution limits may affect performance

4. cloud based ETL/ELT Pipeline for multinational company....

An ELT/ETL pipeline combines data from sales, finance, manufacturing and customer supports into a centralized analytics problem

ELT/ETL :- Extract data from sales Finance and manufacturing systems clean it, and load it into a cloud ware house

Data quality :- validation removes errors and duplicates, while metadata and governance ensure consistency & security

Orchestration :- Apache, Airflow schedules pipelines and handles failures.

cost optimization :- compression, partitioning and query optimization improve performance and reduce costs

5) cloud based ML for delivery prediction..

An logistics company can use cloud based machine learning to predict delivery delays using historical shipments, weather information and live - GPS data

Architecture :- collects GPS, weather, and shipment data and store it in the cloud for analysis

Bigdata technologies :- kafka collects live data, while spark processes large data sets for model training

AIML :-
cloud performs train deploy and monitor prediction models.

model updates :-
Automated data processing and retaining maintain prediction accuracy.